ASP.Net Core settings

Overview



launchsettings.json file

appsettings.json file

Launchsettings.json file

- You will find this file in the "Properties" folder in the project root folder.
- The settings in this file are used when we run this ASP.NET core project either from Visual Studio or by using .NET Core CLI.
- This file is only used on local development machine.
- We do not need it for publishing our asp.net core application.
- If there are certain settings that you want your asp.net core application to use when you publish and deploy your app, store them in **appsettings.json** file. We usually store our application configuration settings in this file.

launchSettings.json

```
{
  "profiles": {
    "http": {
      "commandName": "Project",
      "launchBrowser": true,
      "environmentVariables": {
        "ASPNETCORE_ENVIRONMENT": "Development"
      },
      "dotnetRunMessages": true,
      "applicationUrl": "http://localhost:5055"
    },
    "IIS Express": {
      "commandName": "IISExpress",
      "launchBrowser": true,
      "environmentVariables": {
        "ASPNETCORE_ENVIRONMENT": "Development"
      }
    }
  },
```

```
"$schema":
"http://json.schemastore.org/launchsettings.json",
  "iisSettings": {
    "windowsAuthentication": false,
    "anonymousAuthentication": true,
    "iisExpress": {
      "applicationUrl": "http://localhost:12201",
      "sslPort": 0
    }
  }
}
```

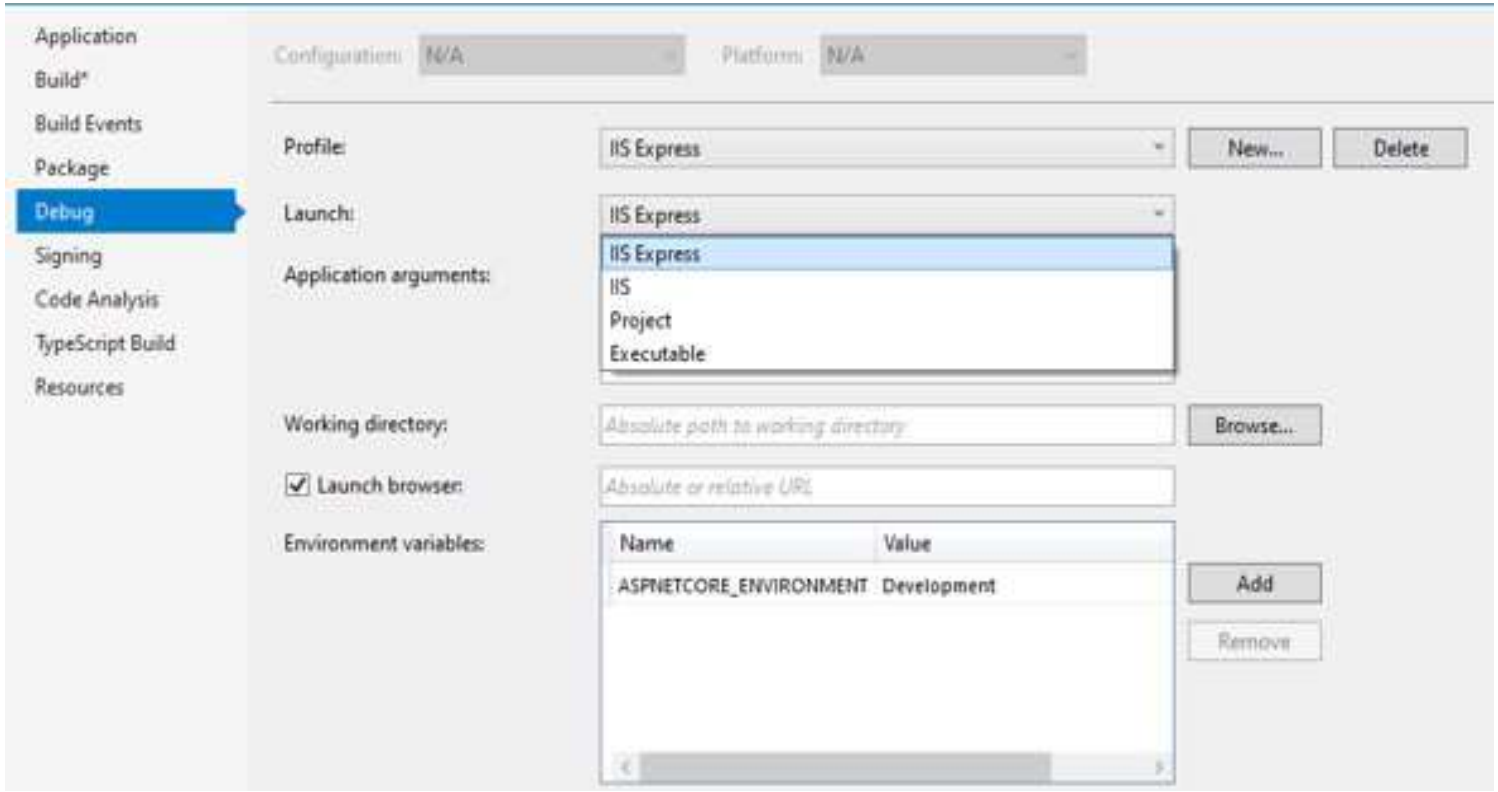
- We have 2 profiles –**IIS/ IIS Express** and **Web Application1**
- When we run the project from Visual Studio by pressing CTRL + F5 or just F5.
- by default, the profile with "commandName": "IISExpress" is used.
- On the other hand, if we run the project using .NET Core CLI (dotnet run), the profile with the "commandName": "Project" is used.
- **Run the specified project with project profile with CLI:**
  dotnet run **--project** WebApplication1.csproj
- **Run the specified project with iisexpress profile with CLI:**
  dotnet run **--iisexpress** WebApplication1.csproj
- **https://docs.microsoft.com/en-us/dotnet/core/tools/**

The value of the **commandName** property can be any one of the following.

- IIS Express
- http

Command Name	AspNetCoreHosting Model	Internal Web Server	External Web Server
Project	Hosting Setting Ignored	Only one web server is used - Kestrel	
IISExpress	InProcess	Only one web server is used - IIS Express	
IISExpress	OutOfProcess	Kestrel	IIS Express
IIS	InProcess	Only one web server is used - IIS	
IIS	OutOfProcess	Kestrel	IIS

- You can change the settings in launchSettings.json file by directly editing the file or we can also change the settings using the Graphical User Interface (GUI) provided by Visual Studio.



appsettings.json file

- We can also have environment specific **appsettings.json** files.
- For example, appsettings.Staging.json for the staging environment.
- In ASP.NET Core, in addition to appsettings.json file, we also have other configuration sources like Environment variables, User Secrets, Command Line Arguments and even our own custom configuration source.

- In ASP.NET Core application configuration settings can come from the

appsettings.json,

appsettings.{Environment}.json

User secrets

Environment variables

Command-line arguments

appsettings.json:

```
{
  "Logging": {
    "LogLevel": {
      "Default": "Information",
      "Microsoft.AspNetCore": "Warning"
    }
  },
  "AllowedHosts": "*",
  "ConnectionStrings": {
    "MyConnection": "server=(local)\\mssqllocaldb;database=mydb"
  }
}
```

Accessing configuration information

```
var builder = WebApplication.CreateBuilder(args);

// Access configuration settings
var mySetting = builder.Configuration["ConnectionStrings:MyConnection"];

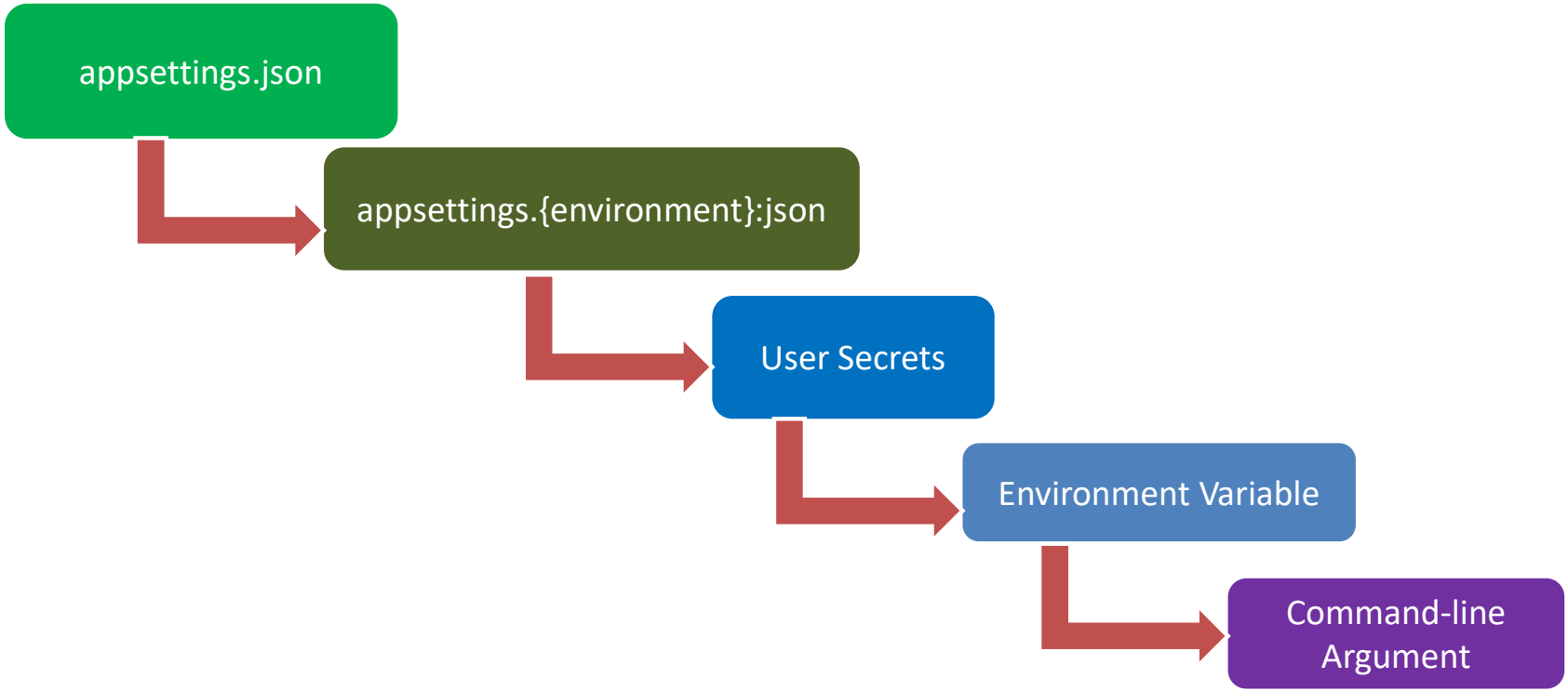

var app = builder.Build();

app.Run(async (context) =>
{
    // Write configuration settings to the HTTP response
    await context.Response.WriteAsync(mySetting);

});

app.Run();
```

The following is the default order in which the various configuration sources are read



You can change this order if you want to or even add your own custom configuration sources in addition to all the existing configuration sources.

References

https://learn.microsoft.com/en-us/aspnet/core/fundamentals/configuration/

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